

## OUTPUTS

### Question 1: Print Numbers from 1 to N

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>javac PrintNumbers.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>java PrintNumbers
Enter a number:
10
1 2 3 4 5 6 7 8 9 10
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>|
```

### Question 2: Print Multiples of 3 between 1 and N

```
1 2 3 4 5 6 7 8 9 10
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>javac MultiplesNumber.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>java MultiplesNumber
Enter a number:
20
3 6 9 12 15 18
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>|
```

### Question 3: Calculate the Factorial of a Number

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>javac FactorialNumber.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>java FactorialNumber
Enter a number:
5
Factorial of 5 is 120
```

### Question 4: Print Even Numbers from 1 to N

```
1 2 3 4 5 6 7 8 9 10
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>javac EvenNumbers.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>java EvenNumbers
Enter a number:
10
2 4 6 8 10
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>|
```

### Question 5: Sum of Odd Numbers between 1 and N

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>java SumOfOddNumbers
Enter a number:
10
The sum of odd numbers from 1 to 10 is : 25
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>|
```

### Question 6: Print All Elements of an Array

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>javac PrintArray.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>java PrintArray
Enter 5 integers:
2 3 4 5 6
2 3 4 5 6
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>
```

### Question 7: Find the Sum of All Elements in an Array

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>javac SumOfArrayElements.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>java SumOfArrayElements
Enter 5 integers:
1 2 3 4 5
The sum of all numbers is: 15
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>
```

### Question 8: Print All Names in a String Array

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>javac PrintStringNames.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>java PrintStringNames
Enter 5 names:
Riddhi Neha Vihan Mayur Anita
Names are: Riddhi
Neha
Vihan
Mayur
Anita
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>
```

### Question 9: Find the Largest Element in an Array

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>javac LargestElemnt.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>java LargestElemnt
Enter 5 integers:
23 34 65 89 67
The largest element is: 89
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>
```

### Question 10: Find the Average of Elements in an Array

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>javac AvgElemnts.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>java AvgElemnts
Enter 5 integers:
10 20 30 40 50
Average of elements = 30.0
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_Building\Assignment_4>
```

### Question 11: Count Positive and Negative Numbers in an Array

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>javac PositiveNegativeInArray.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>java PositiveNegativeInArray
Enter 6 intergers:
-2 5 -8 6 9 3
Positive numbers: 4
Negative numbers: 2
```

### Question 12: Sort an Array in Ascending Order

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>javac SortArray.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>java SortArray
Enter 5 intergers:
53 6 12 34 89
Sorted arrays:
6 12 34 53 89
```

### Question 13: Check if an Array Contains a Specific Element

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>javac SearchElements.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>java SearchElements
Enter 5 intergers:
2 49 54 60 23
Enter the number to search: 60
Found
```

### Question 14: Find the Index of an Element in an Array

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>javac SearchIndex.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>java SearchIndex
Enter 5 intergers:
5 10 20 25 30
Enter the number to search: 20
The number 20 is found at index 2
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>
```

### Question 15: Write a program to print the following pattern:

```
1
2*2
3*3*3
4*4*4*4
5*5*5*5*5
5*5*5*5*5
4*4*4*4
3*3*3
2*2
```

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_C_Building\Assignment_4>javac NumberPattern.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_C_Building\Assignment_4>java NumberPattern
1
2*2
3*3*3
4*4*4*4
5*5*5*5*5
5*5*5*5*5
4*4*4*4
3*3*3
2*2
```

**Question 16: Write a program to print the following pattern:**

```
1
1*2
1*2*3
1*2*3*4
1*2*3*4*5
```

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_C_Building\Assignment_4>javac NumberPattern1.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_C_Building\Assignment_4>java NumberPattern1
1
1*2
1*2*3
1*2*3*4
1*2*3*4*5
```

**Question 17: Write a program to print the following pattern:**

```
1
1*3
1*3*5
1*3*5*7
1*3*5*7*9
```

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_C_Building\Assignment_4>javac NumberPattern2.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi_C_Building\Assignment_4>java NumberPattern2
1
1*3
1*3*5
1*3*5*7
1*3*5*7*9
```

**Question 18: Write a program to print the following pattern:**

```
11111
22222
33333
44444
55555
```

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi Building\Assignment_4>javac NumberPattern3.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi Building\Assignment_4>java NumberPattern3
11111
22222
33333
44444
55555
```

55555

**Question 19: Write a program to print the following pattern:**

```
1
22
333
4444
55555
```

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi Building\Assignment_4>javac NumberPattern4.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi Building\Assignment_4>java NumberPattern4
1
22
333
4444
55555
```

**Question 20: Write a program to print the following pattern:**

```
1
12
123
1234
12345
```

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi Building\Assignment_4>javac NumberPattern5.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi Building\Assignment_4>java NumberPattern5
1
12
123
1234
12345
```

**Question 22: Write a program to print the following pattern:**

```
*****
*   *
*   *
*   *
*   *
*****
```

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi Building\Assignment_4>javac StarPattern.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logi Building\Assignment_4>java StarPattern
*****
*   *
*   *
*   *
*   *
*****
```

**Question 23: Write a program to print the following pattern:**

```
  *
 ***
*****
*****
*****
*****
*****
*****
*****
*****
*****
***
  *
```

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>javac DiamondPattern.java
```

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>java DiamondPattern
```

```
  *
 ***
*****
*****
*****
*****
*****
*****
*****
*****
*****
***
  *
```

**Question 24: Reverse a String**

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>javac ReverseString.java
```

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>java ReverseString
```

Enter a string:

pranali

Reversed string: ilanarp

**Question 25: Count Vowels in a String**

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>javac CountVowels.java
```

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>java CountVowels
```

Enter a string:

programming

The number of vowels in 'programming' is: 3

**Question 26: Check if a String is a Palindrome**

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>javac Palindrome.java
```

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>java Palindrome
```

Enter a string:

madam

The String 'madam' is a palindrome.

### Question 27: String Literal and Object Creation

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>javac StringLiteral.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>java StringLiteral
Both variables point to the same object.
```

### Question 28: String Creation with new Keyword

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>javac StringWithNew.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>java StringWithNew
Using == : false
Using equals() : true
```

### Question 29: String Concatenation and Object Creation

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>javac StringConcatenation.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>java StringConcatenation
str3: helloworld
Is str3 pointing to the same object as str1? false
```

### Question 30: String Pool with intern() Method

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>javac StringIntern.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>java StringIntern
Is str2 and str3 pointing to the same object? true
```

### Question 31: Multiple String Literals with Same Content

```
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>javac MultiStringPool.java
C:\Users\PRANALI\Desktop\Assignment_Solution\Logics_Building\Assignment_4>java MultiStringPool
All strings point to the same object: true
```